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EXPERIMENT 3

Zero-shot, One-shot and Few-shot Prompts - Summarize an Article in 50 Words

AIM

To design zero-shot, one-shot and few-shot prompts for summarizing an article into 50 words and to compare the generated summaries on accuracy, completeness and readability.

PROCEDURE

1. Open an LLM chat interface (such as ChatGPT, Gemini or Claude) in a new chat.
2. Zero-shot: enter an instruction that describes only the task (summarize the given article in 50 words) without any example, and note the response.
3. One-shot: start a new chat and enter the same instruction together with ONE worked example, followed by the actual task. Note the response.
4. Few-shot: start a new chat and enter the same instruction together with TWO worked examples, followed by the actual task. Note the response.
5. Capture a screenshot of each prompt with its generated output.
6. Compare the three outputs and record the observations.

PROMPTS (CODE)

Zero-shot prompt:

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Summarize the following article in 50 words.

Article:
India's electric vehicle market is growing faster than many analysts expected. Sales of electric two-wheelers and three-wheelers have risen sharply over the past three years, helped by government incentives such as the FAME scheme and state-level subsidies on road tax and registration. Falling battery prices have also made electric models more affordable for first-time buyers. Major manufacturers, including Tata Motors, Mahindra and several start-ups, are launching new models and investing in local battery production to reduce dependence on imports.

However, the transition faces serious obstacles. Charging infrastructure remains limited outside large cities, and many apartment dwellers cannot install home chargers. Buyers also worry about driving range, battery life and resale value. In addition, most of India's electricity is still generated from coal, which reduces the environmental benefit of electric transport unless the grid becomes cleaner.

To overcome these barriers, the government has announced plans to expand public charging stations along highways and to encourage battery swapping, which lets drivers replace an empty battery within minutes. Experts believe that if charging networks improve and renewable energy capacity keeps growing, electric vehicles could account for a large share of new sales by 2030, cutting fuel imports and urban air pollution.
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One-shot prompt:

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Summarize the article in exactly 50 words, following the example.

Example article:
Remote work has become common since 2020. Employees save commuting time and
report better flexibility, while companies reduce office costs. However, some
workers feel isolated and managers struggle to measure productivity. Many firms
now adopt a hybrid model, combining office days with remote days to balance
collaboration and freedom.
Example summary:
Remote work offers flexibility and cost savings but brings isolation and
productivity-tracking challenges. Many companies therefore prefer hybrid models
that balance collaboration with freedom.

Article:
India's electric vehicle market is growing faster than many analysts expected.
Sales of electric two-wheelers and three-wheelers have risen sharply over the
past three years, helped by government incentives such as the FAME scheme and
state-level subsidies on road tax and registration. Falling battery prices have
also made electric models more affordable for first-time buyers. Major
manufacturers, including Tata Motors, Mahindra and several start-ups, are
launching new models and investing in local battery production to reduce
dependence on imports.

However, the transition faces serious obstacles. Charging infrastructure remains
limited outside large cities, and many apartment dwellers cannot install home
chargers. Buyers also worry about driving range, battery life and resale value.
In addition, most of India's electricity is still generated from coal, which
reduces the environmental benefit of electric transport unless the grid becomes
cleaner.

To overcome these barriers, the government has announced plans to expand public
charging stations along highways and to encourage battery swapping, which lets
drivers replace an empty battery within minutes. Experts believe that if charging
networks improve and renewable energy capacity keeps growing, electric vehicles
could account for a large share of new sales by 2030, cutting fuel imports and
urban air pollution.
Summary:
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Few-shot prompt:

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Summarize the article in exactly 50 words, following the examples.

Example article 1:
Remote work has become common since 2020. Employees save commuting time and
report better flexibility, while companies reduce office costs. However, some
workers feel isolated and managers struggle to measure productivity. Many firms
now adopt a hybrid model, combining office days with remote days to balance
collaboration and freedom.
Summary 1:
Remote work offers flexibility and cost savings but brings isolation and
productivity-tracking challenges. Many companies therefore prefer hybrid models
that balance collaboration with freedom.

Example article 2:
Plastic pollution is choking oceans, harming marine life and entering the food
chain. Governments are banning single-use plastics, while companies test
biodegradable packaging. Experts say real progress needs better recycling systems
and changes in consumer habits.
Summary 2:
Plastic pollution threatens oceans and food chains. Bans and biodegradable
packaging help, but experts say stronger recycling and changed consumer habits
are essential for real progress.

Article:
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India's electric vehicle market is growing faster than many analysts expected. Sales of electric two-wheelers and three-wheelers have risen sharply over the past three years, helped by government incentives such as the FAME scheme and state-level subsidies on road tax and registration. Falling battery prices have also made electric models more affordable for first-time buyers. Major manufacturers, including Tata Motors, Mahindra and several start-ups, are launching new models and investing in local battery production to reduce dependence on imports.

However, the transition faces serious obstacles. Charging infrastructure remains limited outside large cities, and many apartment dwellers cannot install home chargers. Buyers also worry about driving range, battery life and resale value. In addition, most of India's electricity is still generated from coal, which reduces the environmental benefit of electric transport unless the grid becomes cleaner.

To overcome these barriers, the government has announced plans to expand public charging stations along highways and to encourage battery swapping, which lets drivers replace an empty battery within minutes. Experts believe that if charging networks improve and renewable energy capacity keeps growing, electric vehicles could account for a large share of new sales by 2030, cutting fuel imports and urban air pollution.
Summary:

OUTPUT

Exp 3 - Zero-shot prompting (sample output)

PROMPT

Summarize the following article in 50 words.

Article:
India's electric vehicle market is growing faster than many analysts expected. Sales of electric two-wheelers and three-wheelers have risen sharply over the past three years, helped by government incentives such as the FAME scheme and state-level subsidies on road tax and registration. Falling battery prices have also made electric models more affordable for first-time buyers. Major manufacturers, including Tata Motors, Mahindra and several start-ups, are launching new models and investing in local battery production to reduce dependence on imports.

However, the transition faces serious obstacles. Charging infrastructure remains limited outside large cities, and many apartment dwellers cannot install home chargers. Buyers also worry about driving range, battery life and resale value. In addition, most of India's electricity is still generated from coal, which reduces the environmental benefit of electric transport unless the grid becomes cleaner.

To overcome these barriers, the government has announced plans to expand public charging stations along highways and to encourage battery swapping, which lets drivers replace an empty battery within minutes. Experts believe that if charging networks improve and renewable energy capacity keeps growing, electric vehicles could account for a large share of new sales by 2030, cutting fuel imports and urban air pollution.

MODEL OUTPUT

India's electric vehicle market is expanding rapidly, driven by government incentives, falling battery prices and new launches from manufacturers such as Tata Motors and Mahindra. However, limited charging infrastructure, range worries, resale concerns and coal-based electricity slow the transition. Plans for highway chargers and battery swapping could help EVs take a large share of sales by 2030.

Figure: Zero-shot prompt and sample summary

Exp 3 - One-shot prompting (sample output)

PROMPT

Summarize the article in exactly 50 words, following the example.

Example article:
Remote work has become common since 2020. Employees save commuting time and report better flexibility, while companies reduce office costs. However, some workers feel isolated and managers struggle to measure productivity. Many firms now adopt a hybrid model, combining office days with remote days to balance collaboration and freedom.
Example summary:
Remote work offers flexibility and cost savings but brings isolation and productivity-tracking challenges. Many companies therefore prefer hybrid models that balance collaboration with freedom.

Article:
India's electric vehicle market is growing faster than many analysts expected. Sales of electric two-wheelers and three-wheelers have risen sharply over the past three years, helped by government incentives such as the FAME scheme and state-level subsidies on road tax and registration. Falling battery prices have also made electric models more affordable for first-time buyers. Major manufacturers, including Tata Motors, Mahindra and several start-ups, are launching new models and investing in local battery production to reduce dependence on imports.

However, the transition faces serious obstacles. Charging infrastructure remains limited outside large cities, and many apartment dwellers cannot install home chargers. Buyers also worry about driving range, battery life and resale value. In addition, most of India's electricity is still generated from coal, which reduces the environmental benefit of electric transport unless the grid becomes cleaner.

To overcome these barriers, the government has announced plans to expand public charging stations along highways and to encourage battery swapping, which lets drivers replace an empty battery within minutes. Experts believe that if charging networks improve and renewable energy capacity keeps growing, electric vehicles could account for a large share of new sales by 2030, cutting fuel imports and urban air pollution.
Summary:

MODEL OUTPUT

India's electric vehicle sales are rising sharply thanks to government subsidies, cheaper batteries and new models. Limited charging stations, range and resale concerns, and coal-heavy electricity remain major obstacles. Highway charging and battery swapping plans, plus cleaner power, could help EVs capture a large share of new sales by 2030.

Figure: One-shot prompt and sample summary

Exp 3 - Few-shot prompting (sample output)

PROMPT

Summarize the article in exactly 50 words, following the examples.

Example article 1:
Remote work has become common since 2020. Employees save commuting time and report better flexibility, while companies reduce office costs. However, some workers feel isolated and managers struggle to measure productivity. Many firms now adopt a hybrid model, combining office days with remote days to balance collaboration and freedom.
Summary 1:
Remote work offers flexibility and cost savings but brings isolation and productivity-tracking challenges. Many companies therefore prefer hybrid models that balance collaboration with freedom.

Example article 2:
Plastic pollution is choking oceans, harming marine life and entering the food chain. Governments are banning single-use plastics, while companies test biodegradable packaging. Experts say real progress needs better recycling systems and changes in consumer habits.
Summary 2:
Plastic pollution threatens oceans and food chains. Bans and biodegradable packaging help, but experts say stronger recycling and changed consumer habits are essential for real progress.

Article:
India's electric vehicle market is growing faster than many analysts expected. Sales of electric two-wheelers and three-wheelers have risen sharply over the past three years, helped by government incentives such as the FAME scheme and state-level subsidies on road tax and registration. Falling battery prices have also made electric models more affordable for first-time buyers. Major manufacturers, including Tata Motors, Mahindra and several start-ups, are launching new models and investing in local battery production to reduce dependence on imports.

However, the transition faces serious obstacles. Charging infrastructure remains limited outside large cities, and many apartment dwellers cannot install home chargers. Buyers also worry about driving range, battery life and resale value. In addition, most of India's electricity is still generated from coal, which reduces the environmental benefit of electric transport unless the grid becomes cleaner.

To overcome these barriers, the government has announced plans to expand public charging stations along highways and to encourage battery swapping, which lets drivers replace an empty battery within minutes. Experts believe that if charging networks improve and renewable energy capacity keeps growing, electric vehicles could account for a large share of new sales by 2030, cutting fuel imports and urban air pollution.
Summary:

MODEL OUTPUT

India's EV market is booming due to government incentives, falling battery costs and new model launches. Weak charging infrastructure, range and resale worries, and coal-based power still hold it back. Expanded highway charging, battery swapping and renewable energy could lift EVs to a large share of new sales by 2030.

Figure: Few-shot prompt and sample summary

Comparison of the generated summaries

Criteria	Zero-shot	One-shot	Few-shot
Word limit (50)	57 words - limit exceeded	50 words - met	50 words - met
Accuracy	All facts correct; nothing added from outside the article	All facts correct	All facts correct
Completeness	Most complete: growth drivers, named manufacturers, barriers, plans and the 2030 outlook	Covers drivers, barriers, plans and outlook; manufacturer names omitted	Covers drivers, barriers, solutions and outlook; manufacturer names omitted
Readability	Clear, but long sentences because of the extra length	Clear and smooth; balanced sentences	Most concise; short, direct sentences

RESULT

The zero-shot summary was accurate and the most complete but ran to 57 words, breaking the 50-word limit. The one-shot and few-shot summaries both met the 50-word limit exactly, stayed accurate, and kept the main points (growth, obstacles, government plans and the 2030 outlook). The few-shot summary was the most compact and readable, while all three were factually faithful to the article.

CONCLUSION

Examples help the model respect length constraints and write compactly. Zero-shot prompting favours completeness at the cost of length, whereas one-shot and few-shot prompting give accurate, readable summaries that meet the word limit, with few-shot being the most consistent.